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Utilization of Granite powder in the Production of Sustainable Fiber Reinforced concrete with Improved Durability

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ABSTRACT:

This research studies the feasibility of utilizing Granite Powder (GP) and Glass Fiber (GF) as eco-friendly materials for improving the strength and durability of concrete. Thorough experimental research was conducted in two stages: the effect of various proportions (0% to 40%) of Granite powder (GP) on the mechanical properties of concrete and the inclusion of different percentages (0% to 2%) of Glass fiber (GF) at the ideal mix proportion of GP. Compressive strength, flexural strength, and split tensile strength of concrete were tested at 7, 14, and 28 days of curing. In this study, 250 specimens consist of cubes, cylinders and prisms were cast for mechanical and durability tests. It was observed that 10% partial replacement with GP achieved a 20 percent increase in 28-day compressive strength owing to the enhanced particle distribution and pozzolanic activity. It was found that the use of 1.5% GF also increased the performance of the concrete, with additional percentages of 15% and 18%

in the compressive and flexural strength, respectively, being obtained by analysing the findings to those of the control mixture. The result of the durability tests indicated that the synergistic action of 1.5% GF and 10% GP had greatly enhanced acid resistance and moisture permeability, which were suitable for wet conditions. The outcome of this research highlighted the efficiency and sustainability of the structural benefits of concrete by using waste material in concrete mixtures.

Keywords: Granite Powder (GP), Glass Fibers (GF), Mechanical Properties, Durability, Sustainable Concrete

1. INTRODUCTION

Concrete needs significant improvement in its overall energy and CO₂ footprint, even though some small advancement can make a big difference [1]. The emissions generated during concrete production have negative environmental impacts, particularly on acidification and eutrophication; with coarse aggregate and fine aggregate materials are the major causes of these emissions [2]. Low energy requirement, limited waste, abundance of resources, durability of material, thermal mass, and recycled contents are the environmental effects of sustainable concrete structures [3]. Thus, the conservation of raw materials is essential to minimize the environmental effects associated with concrete production. [4]. Existing studies on granite powder mainly address strength properties, while limited research examines its combined use of glass fiber and its influence on long term durability.

For instance, small gains in efficiency within the cement and concrete value chain can realize up to 50% reductions in greenhouse gas emissions using supplementary cementitious materials and minimising clinker content in cement [5]. Replacement with supplementary cementitious materials, including volcanic material, fly ash, glass, and crop residue products, can significantly decrease the carbon footprint of concrete projects. Another sustainable approach involves the utilization of industrial and agricultural waste products as alternative cementitious materials, such as ground granulated blast furnace slag, silica fume, and

fly ash [6-7]. The use of waste-derived materials including plastics, glass, fly ash, slag, and building demolition residues has been widely recognized for promoting environmental sustainability by reducing waste generation and dependence on virgin raw materials [8-9].

Recent studies, such as those by Olaiya et al. [10-12], have demonstrated that agro-industrial by-products like banana leaf ash and other pozzolanic wastes can effectively serve as sustainable raw materials in producing eco-friendly concrete and sandcrete bricks, achieving mechanical performance comparable to conventional materials while reducing emissions. Similarly, Olaiya et al. [13] emphasized the potential of sawdust composites as viable alternatives in the construction industry, highlighting their contribution to circular economy practices. Furthermore, Olaiya et al. [14] explored sustainable building practices in modern laboratories, reinforcing the role of resource-efficient design and material reuse in advancing sustainable infrastructure development. Collectively, these studies affirm the effectiveness of waste valorization strategies in reducing carbon intensity and supporting green construction initiatives.

The recycling of marble and granite residues has also gained attention for its financial and ecological benefits, as these materials reduce natural resource consumption and promote a circular materials economy [15]. The construction cost can be reduced by approximately 0.54% to 2.1% if waste granite is recycled, accompanied by a 5% reduction in CO₂ emissions, while maintaining equivalent or improved mechanical properties compared to conventional materials [16]. Incorporation of granite powder as a filler enhances packing density, reduces cement consumption, and strengthens the mortar microstructure [17]. Partial cement replacement of up to 20% using granite waste dust improves resistance to acid and chloride attack without compromising structural integrity [18]. In mass-produced sand concrete, granite powder substitution enhances mechanical strength by up to 20.9% while reducing autogenous shrinkage by 19.7% [19]. However, excessive inclusion of granite dust can result in unsatisfactory performance due to reduced binder content and increased porosity [20]. Controlled partial replacement typically below 15% of fine

aggregate content has been found beneficial, improving strength, durability, and resistance to sulphate attack when combined with chemical treatment [21-22].

Utilization of granite waste as coarse aggregate improves fresh characteristics and mechanical behaviour of hardened concrete in high-performance self-compacting concrete [23]. Hardened concrete is tough in compression but fragile in tension; reinforcements are introduced to overcome the limitation [24]. The slump reduction of the altered mixtures was attributed to the glass fibers' bridging effect [25]. Recycled glass fibers in concrete are alternatives that take green technology further by replacing cement with old materials, generating less waste, and improving mechanical performance to achieve sustainable construction [26]. Glass fibers in concrete are likely to withstand transport properties, chemical attack, and freeze/thaw action in cold climates beyond mere resistance to pressure bearing and thereby enhance durability [27]. Glass fiber can be employed at lower proportions of concrete to enhance flexural strength with a low slump value without decreasing workability and compressive strength [28]. Glass fibers added to the concrete enhance its strength through a slight improvement in compressive strength, flexural strength, and split tensile strength and a decrease in bleeding and permeability [29]. The incorporation of 1.0% glass fiber in concrete enhances its compressive strength by approximately 13.69% compared to ordinary concrete [30]. Although plastic waste tends to reduce the strength properties of concrete, additives such as fly ash and silica fume have been used to counteract this effect and improve performance [31].

In general, lower water-to-binder ratios lead to a better mechanical performance of the glass fiber-reinforced concrete with compressive strengths up to 160 MPa if micro-glass fibers between 1.5% and 3% are added [32]. Using 1.25% glass fiber within concrete resulted in increased mechanical strength, optimum embodied carbon value, and improved impact energy [33]. These fibers enhance the concrete's strength and toughness but diminish its usability. The ideal dose of 2.0% is recommended [34]. It has been noticed that the functioning of granite

powder and glass powder stabilization of Black cotton soil was greater than that based only on glass powder ^[35]. It reduces environmental impact and improves sustainability by exploiting industrial by-products in production ^[36]. Compressive strength and porosity values of marble waste powder and glass fiber reinforced cement mortars enhanced substantially at high temperatures, and a reasonable agreement is present between the experimental and statistical results ^[37].

2. MATERIALS:

The specially formulated and utilized concrete mix used in this research was adequately composed by Ordinary Portland Cement (OPC), fine aggregate, coarse aggregate, Granite powder, glass fibers, and water, with proper strength and durability. All the materials were readily available in the Tirupur industrial belt, Tamil Nadu, and the ease of availability facilitated easy procurement with normal material quality. The physical characteristics of the aggregates are presented in Table 1, as per the requirements of IS 2386 and BS 882, and they qualify for high-performance concrete applications. As a goodwill gesture toward green construction and sustainability, Granite powder was utilized as an alternative for cement with fractions of 10%, 20%, 30%, and 40%. A detailed study of its chemical oxide content, as shown in Table 2, reveals the prevalence of alumina and silica, which are heavily involved in secondary hydration reactions, thereby contributing to the improved microstructural cohesion of the hardened cementitious matrix. Apart from the mechanical behaviour of the concrete, glass fibers were incorporated by volume percentages of 0.5%, 1%, 1.5%, and 2%. These 5–15 μm diameter fibers of high tensile strength with 10–50 mm length possess an excellent tensile strength of 2 to 3 GPa. Table 3 shows the unique characteristics of the glass fibers used in this study. This well-selected set of materials, complemented by their experimentally confirmed characteristics, provides a comprehensive and technically robust foundation for creating high-strength concrete, offering the optimal combination of strength, durability, and environmental acceptability.

Table 1: Physical Properties of Aggregates

Properties	Granite Powder	Fine Aggregate	Coarse Aggregate
Specific Gravity	2.82	2.6	2.7
Fineness Modulus	3.58	2.25	5.95
Water Absorption (%)	0.78	0.86	1.0

Table 2: Granite powder Properties

Specifications	Range
Fineness (Passing 75 μ m)	85 – 95%
Specific Gravity	2.82
Water Absorption (%)	0.78
Loss on Ignition (%)	0.5 – 1.5
Silicon Dioxide (SiO ₂)	65 – 75%
Aluminum Oxide (Al ₂ O ₃)	12 – 18%
Calcium Oxide (CaO)	2 – 5%

Table 3: Glass Fiber Properties

Specifications	Range
Diameter (mm)	0.5
Length (mm)	25
Aspect Ratio	50

Tensile Strength (MPa)	1000 - 4000
Young's Modulus (GPa)	70 - 80
Density (g/cm ³)	2.6 - 2.7

3. MIX DESIGN

The concrete mix grades of M25 were prepared as per the requirements of IS 10262:2019. The final mix ratio was 1:1.21:2.91 cement, fine aggregate, and coarse aggregate. A water-cement ratio of 0.46 was implemented to fulfil the requirements of strength and workability. An optimised mix design was used to obtain a design for a target mean strength higher than the 25 MPa characteristic strength and provide sufficient performance of the concrete.

4. METHODS

Experimental studies were conducted in two phases to solve the mechanical and durability problems of concrete by blending Granite powder and glass fibers. The test specimens were shaped into (150 × 150 × 150 mm) cubes, cylinders (150 mm diameter × 300 mm length), and prisms (100 × 100 × 500 mm). In the tests, cement was substituted with Granite powder in varying percentages of 10%, 20%, 30%, and 40% of the total weight of each specimen, with the aim of investigating their mechanical properties. With slump cone and compaction factor tests, the Workability of Concrete were evaluated. The mechanical strength tests, compressive strength, split tensile strength and flexural strength were conducted as per IS 516, while the durability studies were carried out in accordance with ASTM standards. The results from testing the compressive strength, split tensile strength, and flexural strength at 7, 14, and 28 days were achieved to enhance the Mechanical characteristics of the concrete. The ideal mix proportion with Granite powder were determined, and the specimens were prepared with varying percentage of glass fibers from 0.5% to 2.0%. To determine durability, water absorption tests were conducted to measure permeability, and acid resistance tests were performed by submerging the samples in 5% H₂SO₄ and 5% HCl

solutions. Weight loss and strength loss tests were also carried out. These comprise a penetration test for chlorides with reference to the RCPT as established by ASTM C1202. In this study, 250 specimens, comprising cubes, cylinders, and prisms, were cast for mechanical and durability tests. All specimens were subjected to a standard water curing by immersion in water tanks maintained at 27 ± 2 °C. These studies together provided a comprehensive knowledge about the performance and the behaviour of concrete incorporating Granite powder and glass fibers exposed to mechanical and environmental conditions.

5. RESULTS AND DISCUSSIONS:

5.1. Influence of Granite powder on mechanical properties of concrete:

The possible strength tests were applied to the specimen for the study of the impact of GP on the mechanical properties of the concrete, and the result achieved is described.

5.1.1. Compressive Strength

The compressive strength of concrete was significantly enhanced by partial cement replacement with Granite powder, with the best results at a 10% level of replacement. The enhancement is primarily due to the finely ground GP's filling effect, which improves particle packing, resulting in a decrease in porosity and an increase in interparticle bonding, thereby enhancing the interlocking of aggregates with the cement paste matrix. This results in a denser and more coherent form.

This is confirmed by Li et al. ^[38], who emphasized the capability of GP to continue to develop the microstructure and close voids. In addition, Ghorbani et al. ^[39] demonstrated that the pozzolanic activity of GP, based on its silica content, facilitates the formation of secondary C-S-H gel, leading to an additional strength gain. According to Ghorbani et al. ^[40], when replacement surpasses 10%, a dilution effect occurs. Excessive GP leads to porosity gain and decreased mechanical performance owing to the reduced cement content, restricted hydration, and strength-reducing compound formation. Figure 1 shows the variation of compressive

strength with various proportions of granite powder alone for 7, 14 and 28 days. The compressive strength of 10% GP replacement was 33.12 MPa at 28 days, corresponding to a 6.2% increase over the control concrete, which confirms the optimum improvement recorded at this replacement percentage level.

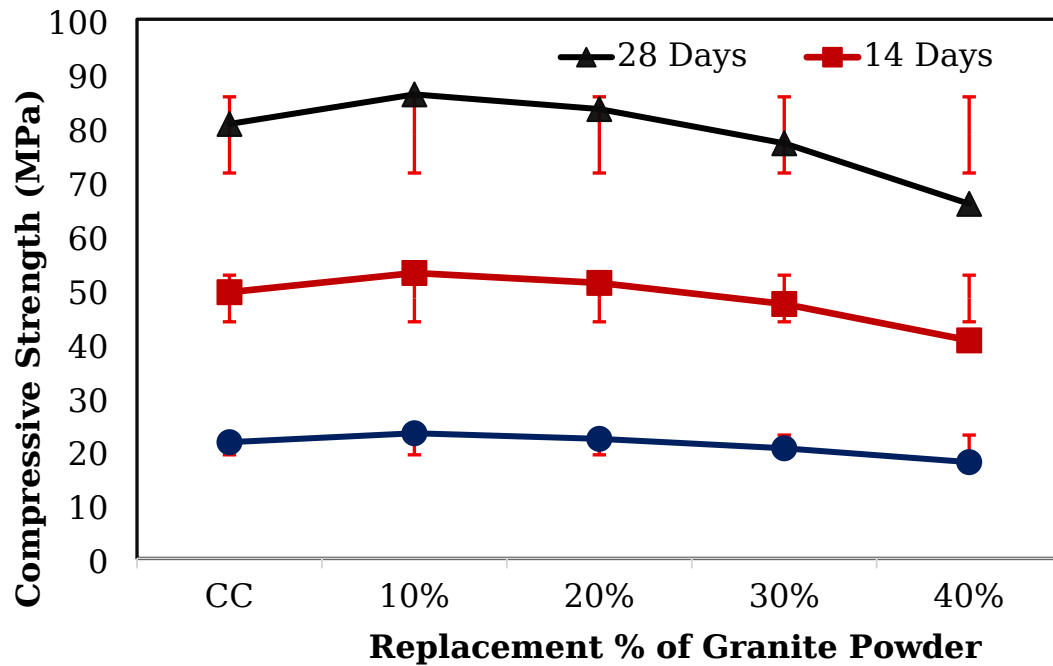


Figure 1: Variation of compressive strength (Mpa) of concrete with different proportions of granite powder (%)

5.1.2. Flexural Strength

The flexural strength of concrete with the addition of Granite powder was significantly enhanced to a level of 10% substitution. Specimens at 28 days of curing with 10% substitution of GP showed a comparative improvement in flexural strength as compared to the control mix. This is appropriate to the denser particle and improved particle gradation obtained by GP, resulting in reduced porosity as well as increased interfacial bonding in the cementitious matrix, which is supported by Vijayalakshmi et al. (2013) [41]. The pozzolanic action of GP is also critical, as it can facilitate further development of the C-S-H gel and enhance the strength of the matrix. All these roles collectively lead to improved bending resistance and cracking initiation retardation, hence increasing the concrete ductility.

When, however, the replacement level surpassed 10%, a progressive loss of flexural strength occurred. This may be attributed to a decrease in the effective cementitious material and impaired interlocking of aggregates, which degrades the efficiency in transferring loads across the matrix. The variation of flexural strength of concrete with varying proportions of granite powder after 7 and 28 days is represented in Figure 2. Following 28 days of curing, the flexural strength for 10% partial replacement of granite powder was 4.42 MPa. When compared to control concrete, there was an increase of 12.8% which aligns with the maximum improvement obtained at this percentage.

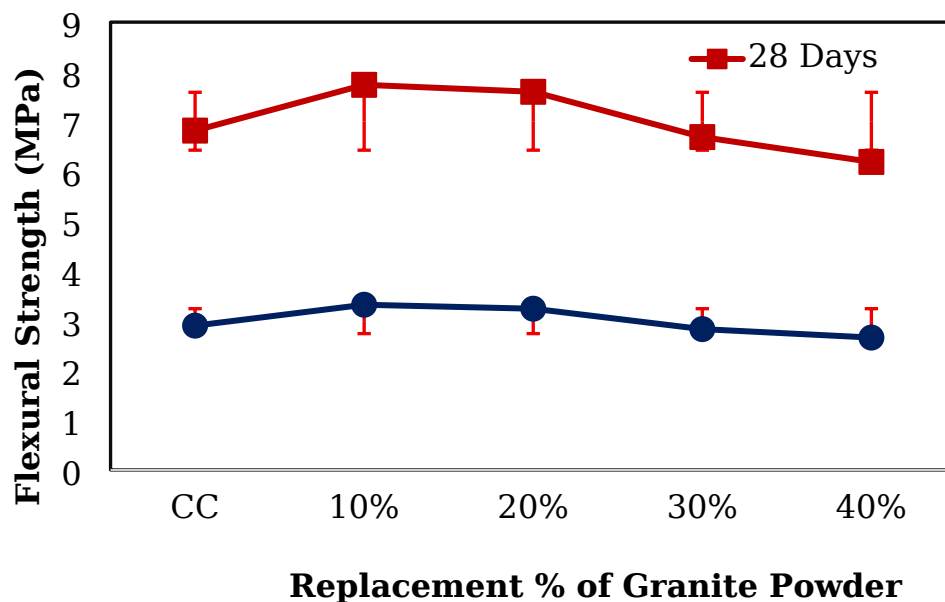


Figure 2: Variation of flexural strength (Mpa) of concrete with different proportions of granite powder (%)

5.1.3. Split Tensile Strength

Concrete with granite powder (GP) exhibited a steady yet modest increase in split tensile strength, particularly at lower replacement percentages. Improvement was observed at 10% replacement of GP with minority due to increased matrix densification and more effective stress transfer, thereby retarding crack extension. This is in line with reports in research publications in which maximized utilization of waste material improved microstructural integrity.

As the level of GP replacement increased above 10%, a gradual loss of tensile strength occurred. The loss of tensile strength is characterized by weaker bonding in the aggregate-matrix interface as well as reduced availability of the cementitious phases for superior interfacial transition zones. This concurs with Tahir et al. [42], who found that a 40% level of GP substitution achieved a tensile strength comparable to that of the control concrete mixture, thereby demonstrating the capability of GP to sustain tensile performance with minimal compromise. This demonstrates the potential of GP to be utilised in sustainable concrete applications without compromising structural integrity. Figure 3 illustrates the variation in tensile strength of concrete incorporating different proportions of granite powder. The split tensile strength of concrete with 10% partial replacement of granite powder was recorded as 3.38 MPa. In comparison with control concrete, an 8% increase was observed in the 10% GP replacement mix.

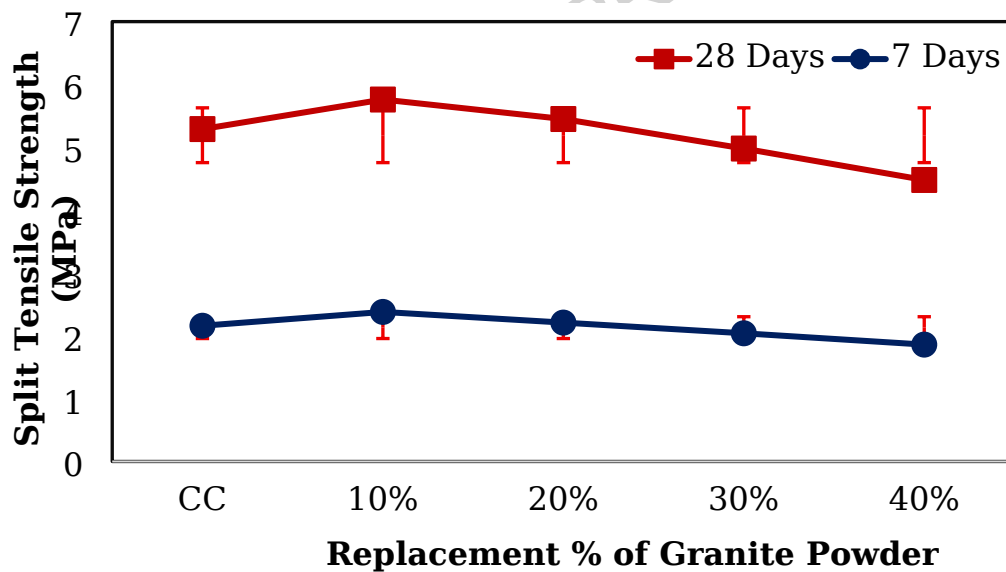


Figure 3: Variation of split tensile strength (MPa) of concrete with different proportions of granite powder (%)

5.2. Influence of 10% Granite Powder and Glass Fiber on the Mechanical Properties of Concrete

5.2.1 Compressive Strength

Figure 4 presents the variation in compressive strength for concrete containing constant 10% granite powder with glass fiber additions ranging from 0% to 2% at 7, 14 and 28 days. Various proportions of granite powder alone for 7, 14 and 28 days. Maximum improvement in strength was noted at 1.5% of GF, and compressive strength was enhanced significantly when compared with the normal control mix. The enhancement is due to a dual mechanism: the dense packing effect of GP improves the microstructure and decreases porosity, while the incorporation of GF effectively regulates the generation of microcracks and facilitates the distribution of internal stress. Matrix densification and crack-bridging action synergised to obtain concrete with increased strength and extended durability. The increase in compressive strength at constant 10% granite powder with 1.5% glass fibers is due to the combined filler effect and reinforcement effects. Granite powder improves particle packing and reduces voids, leading to a denser microstructure and enhanced C-S-H formation. Glass fibers control micro crack propagation and improve stress distribution under load.

Despite that, the compressive strength decreases with fiber content above 1.5% of GF, perhaps due to fiber clumping and reduced workability, highlighting the significance of the optimal fiber content. Generally, the inclusion of a 10% GP and 1.5% GF mix yielded the best combination of sustainability and mechanical properties, marking a new path towards the future for sustainable high-strength concrete. The optimum blend of 10% granite powder (GP) with 1.5% GF developed a compressive strength of 36.4 MPa, which registered a 16.7% gain compared to the control concrete.

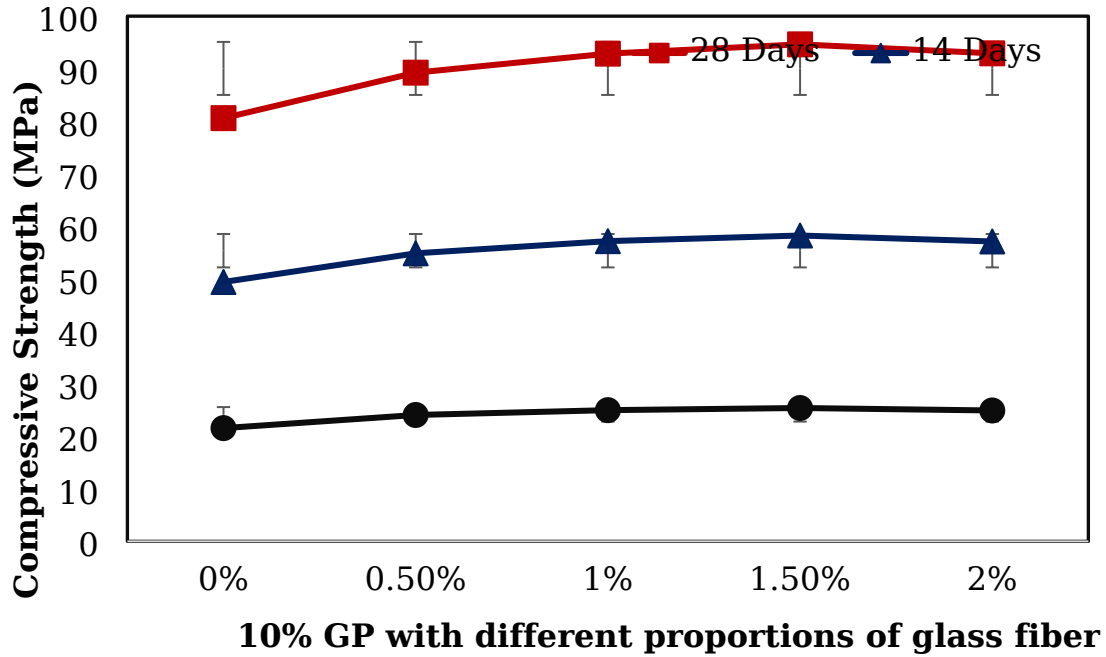


Figure 4: Results of the compressive strength (Mpa) test with a constant 10% GP with varying proportions of glass fiber (%)

5.2.2 Flexural Strength

Figure 5 shows the variation in flexural strength for concrete containing constant 10% granite powder with glass fibre additions ranging from 0% to 2% at 7, 14 and 28 days, and various proportions of granite powder alone for 7 days and 28 days. In comparison with a fiber-free control mix, 28 days showed an enhancement in flexural strength, when there was 1.5% GF and 10% GP in a specimen. Moreover, the addition of GP enhanced matrix densification, thus enhancing the resistance of the concrete to flexural stresses.

However, even at over 1.5% GF content, a trend towards declining flexural strength was observed. This would most likely be an effect of the fibers agglomeration and higher voids, inclining to impair matrix continuity and effectiveness in load transfer. Thus, a 1.5% dosage of GF combined with a 10% dosage of GP represents an optimum level in both improving crack resistance and enhancing flexural strength, without compromising overall structural integrity in concrete. The mix obtained with 10% GP and 1.5% GF achieved a flexural strength of 4.24 MPa, representing an 8.2%

increase compared to the control mix. The increase in flexural strength at constant 10% granite powder with 1.5% glass fibers is due to a combined effect of matrix densification and fiber reinforcement. Granite powder acts as a micro-filler, improving particle packing and refining the pore structure, thereby strengthening the cement matrix and the interfacial transition zone. The glass fibre bridge microcracks and delays their propagation, converting brittle behaviour into a more ductile response under bending. The interaction enhances load transfer, resulting in higher flexural strength compared to conventional concrete.

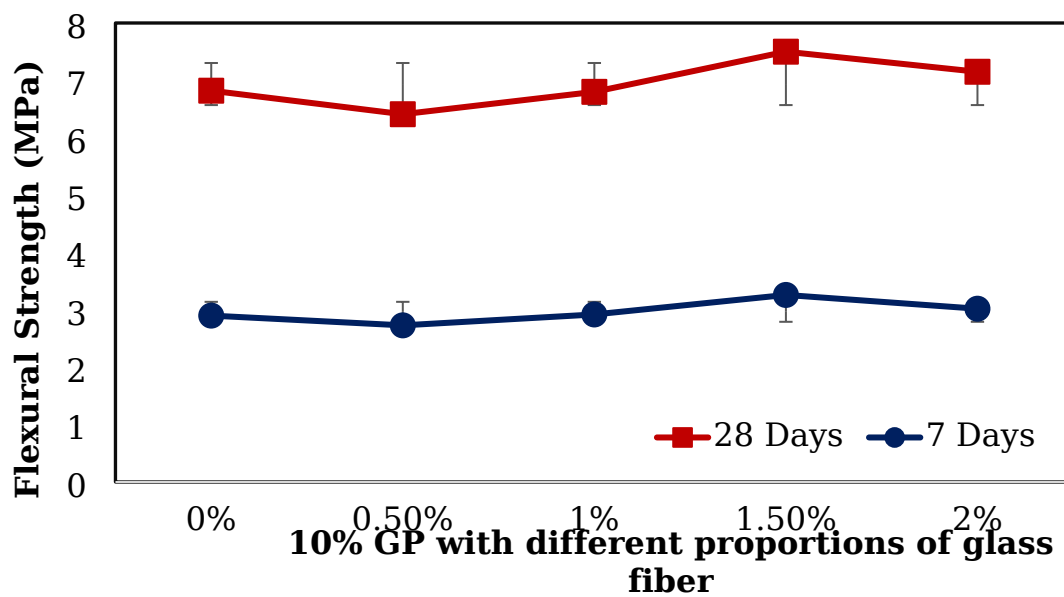


Figure 5: Results of the flexural strength (MPa) test with a constant 10% GP with varying proportions of glass fiber (%)

5.2.3 Split tensile Strength

Figure 6 presents the outcomes of the split tensile strength test on concrete specimens containing 10% GP and different glass fibre (GF) content ranging from 0-2%, tested after water curing for 7, 14, and 28 days. The split tensile strength of the 1.5% GF and 10% GP specimens increased in strength after 28 days in comparison with the control mix. The improvement is due to the synergistic interaction between GP and GF. The filler role of GP enhances densification of the matrix, and glass fibers bridge microcracks, transfer tensile stress, and retard crack growth. Upon the increase in the fiber content to more than 1.5%, a progressive

reduction in tensile performance due to fiber agglomeration, loss of workability, and development of voids in the matrix was observed.

Generally, the 10% GP and 1.5% GF blend provided the best compromise, increasing tensile strength without reducing matrix strength. The above remarks emphasize that the modest addition of fiber, with or without GP, improves durability and resistance to fracture and makes the concrete more resistant to tensile stress. A 26.5% increase was noted in the strength of concrete under split tensile with 10% GP and 1.5% GF. The fine particles of granite powder enhance matrix density by improving pore packing and strengthening the interfacial transition zone. Glass fibres, on the other hand, act as crack-bridging elements, resisting crack propagation and improving post-cracking behaviour. The denser matrix formed by granite powder enhances fiber - matrix bonding, allowing the fibers to transfer tensile stresses more effectively. The interaction results in better crack resistance and improved tensile load carrying capacity, leading to higher split tensile strength.

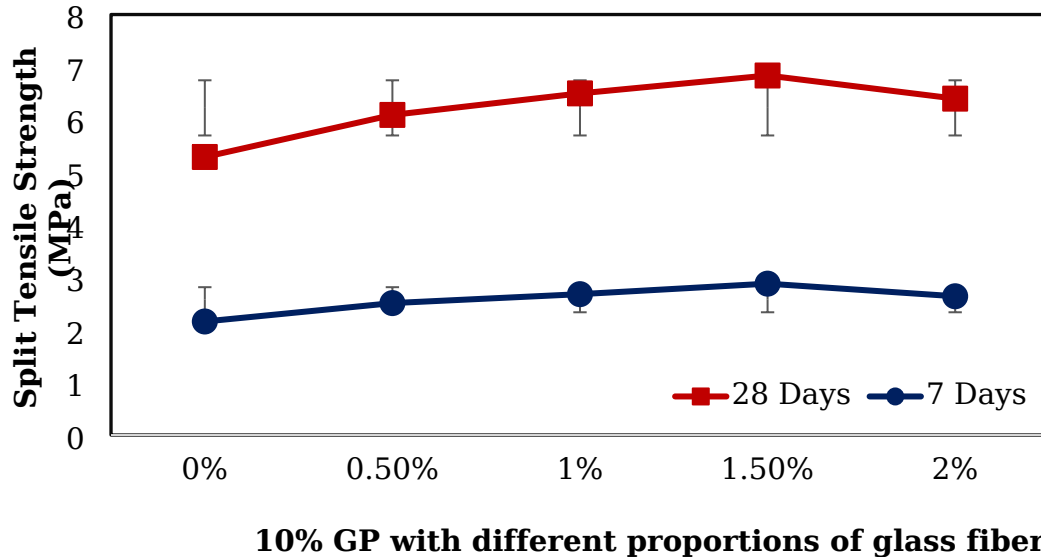


Figure 6: Results of the split tensile strength (Mpa) test with a constant 10% GP with varying proportions of glass fiber (%)

5.3. Durability studies

Durability performance of concrete containing 10% Granite powder (GP) and different glass fiber (GF) contents was tested under acid attack conditions.

5.3.1. Loss in Strength due to acid attack

Table 4 presents the strength loss percentage of concrete after exposure of HCl and H₂SO₄. At 28 days of exposure, the conventional concrete showed significant compressive strength losses owing to acid-induced leaching and matrix degradation. The addition of 10% GP diminished the loss in compressive strength by about 4.5% in both HCl and H₂SO₄ compared to the control mix. This is mainly due to attributed to the pozzolanic effect and micro-filler action of granite powder that densifies the interfacial transition zone and reduces permeability. Figures 7 and 8 shows the reduction in strength of concrete specimens subjected to HCl and H₂SO₄, respectively.

Table 4: Results of strength loss (%) of concrete after exposure to HCl and H₂SO₄ solutions

Mix ID	Strength loss due to HCl (%)	Strength loss due to H ₂ SO ₄ (%)
CC	14.86	20.68
10% GP	10.32	16.12
10% GP + 1.5 % GF	7.86	12.48

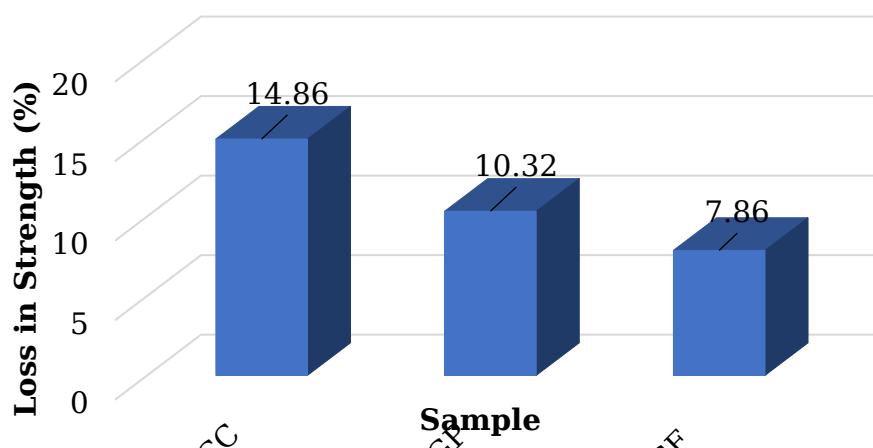
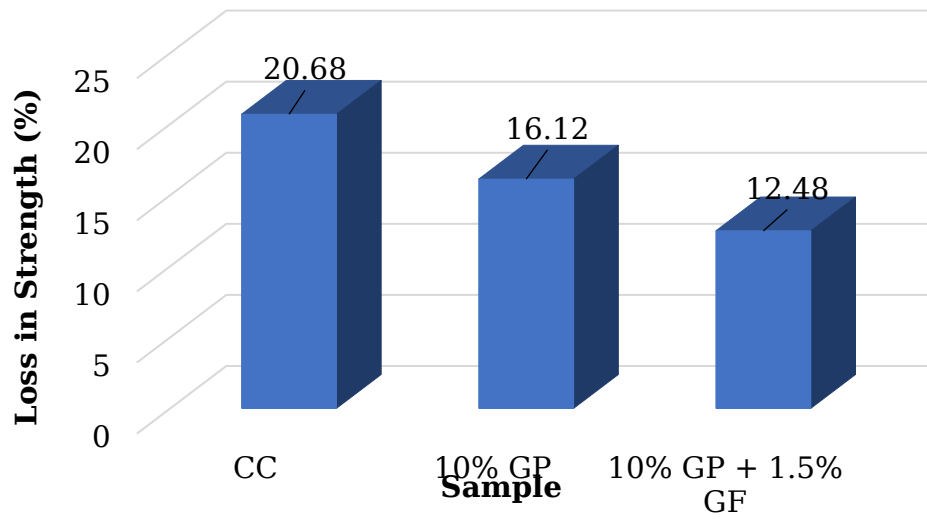


Figure 7. Loss (%) in Strength due to HCL**Figure 8:** Loss in Strength due to H_2SO_4

The incorporation of 1.5% GF in combination with 10% GP also increased resistance to acid attack, with losses in strength 7.0% and 8.2% less than the control mix when exposed to HCl and H_2SO_4 , respectively. Glass fibers are primarily effective through crack-bridging mechanisms that enhance tensile strain capacity and restrain crack extension, thereby preserving composite integrity against acid-induced microcracking. The fibers also minimize channels for acid penetration and slow down the degradation of the cementitious matrix. Indeed, sulfuric acid was more aggressively deteriorating due to gypsum formation and expansive reaction, but the GP+GF specimens still showed significantly improved strength retention.

5.3.2. Loss in Weight due to acid attack

Table 5 shows the weight loss percentage of concrete after exposure of HCl and H_2SO_4 . The control specimens without additives had the maximum weight loss under both the acid exposures in HCl and H_2SO_4 , indicating extensive degradation of the cementitious matrix due to acid attack. Adding 10% granite powder as a partial substitution of cement lowered the weight loss to 1.04% for HCl and 2.36% for H_2SO_4 immersion compared to the concrete mix. The inclusion of 1.5% glass fibers together with granite powder also improved durability, reducing weight losses to

1.68% in HCl and 2.94% in H_2SO_4 when compared with the control mix. These observations establish that granite powder replacement of cement considerably improves acid resistance by microstructure densification and reduction in porous penetrability, resisting acid intrusion and leaching of vulnerable phases like calcium hydroxide. The combined application of glass fibers provides mechanical reinforcement by micro-crack bridging, limiting crack extension under chemical stress, and thus maintaining matrix integrity on acid exposure. Figures 9 and 10 show the loss of weight of concrete specimens exposed to hydrochloric acid (HCl) and sulfuric acid (H_2SO_4), respectively.

Table 5: Results of weight loss (%) of concrete after exposure to HCl and H_2SO_4 solutions

Mix ID	Weight loss due to HCl (%)	Weight loss due to H_2SO_4 (%)
CC	3.2	5.6
10% GP	2.16	3.24
10% GP + 1.5 % GF	1.52	2.66

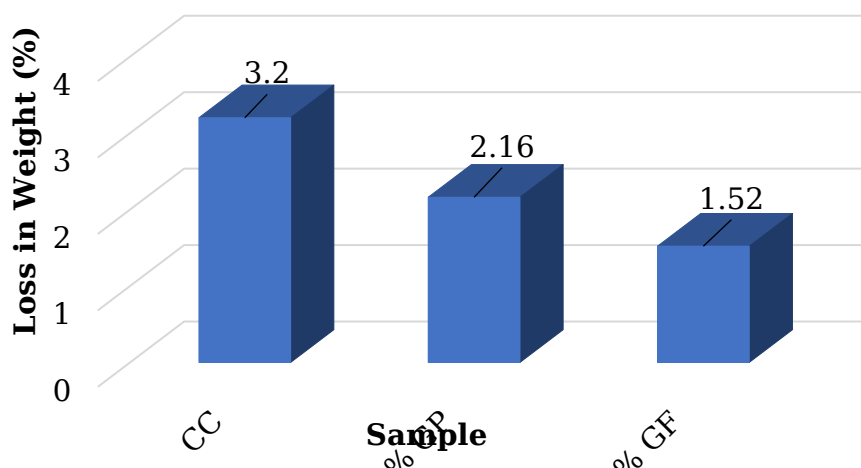


Figure 9: Loss in Weight due to HCL

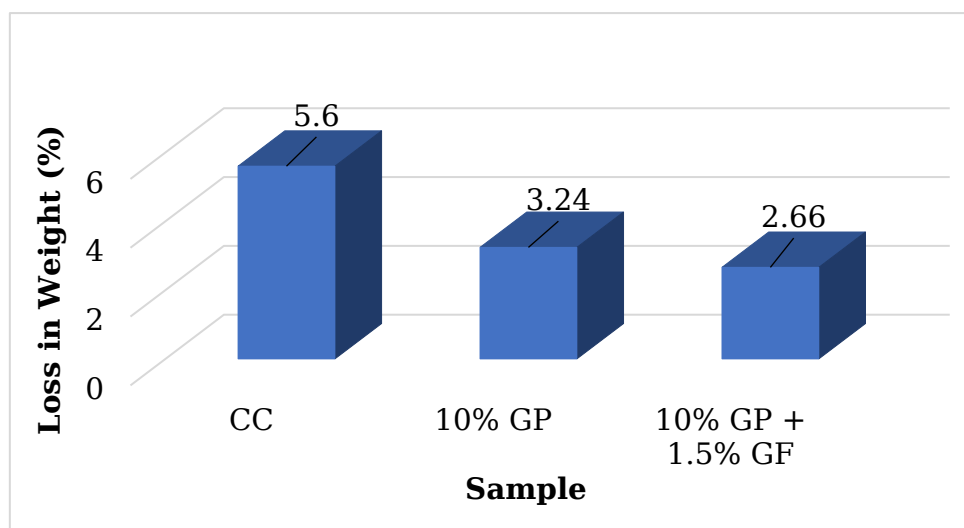


Figure 10. Loss in Weight due to H_2SO_4

The loss of weight due to sulfuric acid attack was greater for all the specimens than in hydrochloric acid. This agrees with the established mechanism of H_2SO_4 degradation, in which the sulfuric acid interacts with hydration products formed expansive ettringite and gypsum phases that lead to surface scaling and spalling. The fiber and granite powder modifications seem to counteract these effects more effectively in sulfuric acid solutions, perhaps because of decreased availability of calcium hydroxide and enhanced microstructural cohesion.

5.3.3. Rapid Chloride Penetration test (RCPT)

Table 6: RCPT results

Concrete sample	Charge Passed (Coulombs)	Chloride permeability
CC	2865.78	Low
10% GP	2532.6	Low
10% GP + 1.5 % GF	2168.2	Low

Rapid Chloride Permeability Test (RCPT) quantifies the resistance of concrete to chloride penetration, an important durability parameter in

aggressive conditions such as sea and road surface applications subject to de-icing salts. A 10% addition of granite powder (GP) maximizes concrete microstructure by minimizing pore connectivity, producing a denser matrix that resists chloride penetration.

In combination with 1.5% glass fibers (GF), the composite exhibits improved performance - GP optimizes the internal structure and GF toughens the material against microcracking, together minimizing charge passage measurements in RCPT tests. This dual adjustment generates concrete with improved chloride resistance while yielding environmental advantages in terms of industrial byproduct use and lower cement usage, thereby well-fitting infrastructure applications demanding durability as well as sustainability.

The synergistic action of GP and GF not only enhanced early-stage resistance but also ensured long-term durability by decreasing crack extension with time. Under long-term exposure, the robust matrix formed because of GP addition remains effective in providing long-term chloride resistance. In general, the incorporation of GP and GF resulted in denser, less permeable concrete with excellent potential for use in aggressive chloride environments.

5.3.4 SEM Analysis of Control, Granite powder and Glass Fiber added concrete

Scanning Electron Microscopy (SEM) is a critical technique applied to evaluate the microstructure of concrete at high resolution and magnification. The SEM micrographs were captured from the fractured surfaces of each sample ^[43-45]. Figure 11 (a) and (b) shows the SEM images of control concrete and granite powder with glass fiber added concrete. SEM analysis helps to ascertain how the additions influence the internal structure of concrete upon addition of glass fibers and granite powder. The well-bonded surfaces between glass fibers and hydration products were visible using SEM. Granite powder is a finely ground by-product and its fineness and particle shape allow it to act as a micro-filler in the cementitious matrix. These micro-fines fill the micro-voids and

capillary pores among large cement grains and aggregates, resulting in a denser and more compact microstructure. This decreases the permeability of the concrete, which in turn improves its resistance to water penetration, chloride penetration, and chemical attack, all of which are key considerations for long-term durability. Although granite powder is mostly inert, it can contain reactive silica and alumina, depending on its mineral content and fineness. These compounds can react with calcium hydroxide (CH), one of the products of cement hydration, to produce calcium silicate hydrate (C-S-H) [46-47], a key component that contributes to the primary strength in the initial phase. When granite powder and glass fiber are combined, they create a concrete matrix that is denser and more resilient, as shown by compact hydration products, well-bonded interfaces, and fewer voids. Additionally, adding fiber enables the concrete to deform considerably before failing.

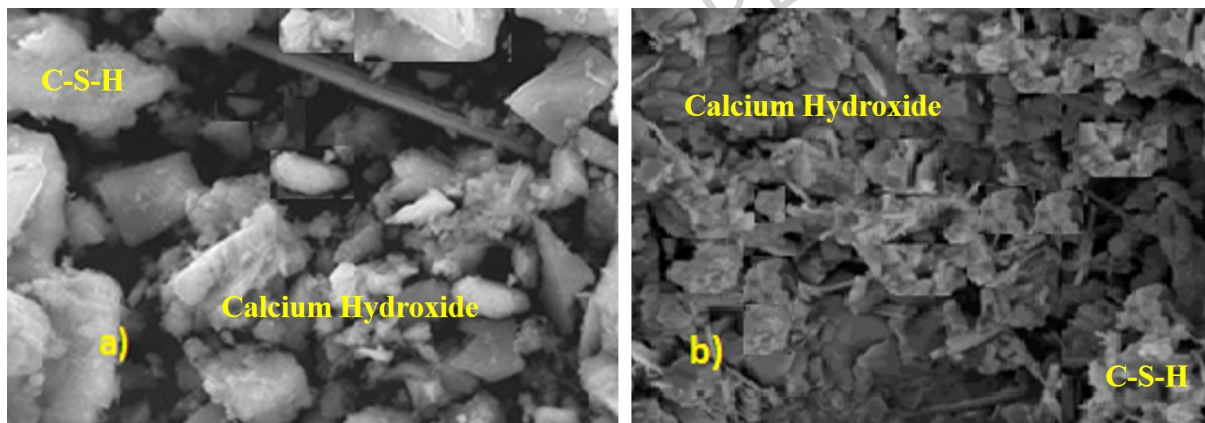


Fig. 11. SEM image of (a) Control Concrete, and (b) Granite Powder and Glass Fibre added concrete

6. CONCLUSION

Increased utilization of Granite powder (GP) and glass fibers (GF) in concrete enhances the mechanical strength and durability of concrete. The optimal proportion of 10% GP and 1.5% glass fibers yielded the following:

1. The Partial replacement of 10% granite powder with cement will enhance the compressive strength, tensile strength, and flexural

strength of concrete. This result indicates the feasibility of using granite powder as a potential alternative in concrete methods.

2. By incorporating 1.5% glass fibers as an addition along with partial replacement of 10% granite powder with cement further enhances compressive strength, split tensile strength and flexural strength parameters were enhanced by concrete mix proportion containing 10% GP + 1.5% GF, which determine the effects of natural fibers to establish mechanical properties.
3. The enhancement results from the synergistic interaction of 10% GP refines the Interfacial transition zone and reduce pore spaces, enhancing matrix densification, while the 1.5% GF bridge cracks and control their propagation. Together, the filler effect of granite powder and reinforcement action of glass fibers produces a denser, stronger, and more durable concrete mix.
4. The concrete samples, which incorporated 10% of granite powder and 1.5% of glass fiber, exhibited improved durability properties when exposed to acidic environmental conditions that leads to increased resistance to chemical degradation. The reduction in strength and increase in weight loss may be attributed to the limited pozzolonic reactivity of granite powder, increased porosity due to cement dilution, reduced workability and compaction issues arising from fiber addition and a weaker fiber interface, all of which makes the concrete more vulnerable to aggressive environments.
5. Concrete with 10% GP showed low permeability, as attested by a charge passed of 1841.36 C. The low permeability classification implies improved durability and lifespan of the concrete under different environmental conditions.
6. Further research is recommended on external durability test such as freeze thaw, carbonation, and in-situ exposure supported by life cycle assessment analysis to substantiate sustainability claims. To strengthen the sustainability framework, future work should include accelerated durability assessments and full life cycle assessment analysis comparing GP-GF concrete with conventional mixes.

Declarations

Conflict of interest

The authors declare that there is no conflict of interest

Ethics, Consent to Participate, and Consent to Publish Declarations

Not applicable

Clinical Trial

Not applicable

Availability of data and materials

All relevant data generated or analyzed during this study are included in the manuscript.

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Authors Contribution

SA: Conceptualization; Formal analysis; Investigation; Methodology; Resources; Software; Validation; Writing—original draft.

SSRD: Conceptualization, Methodology, Investigation, Validation, Writing—original draft, Writing - review and editing

GKA: Conceptualization, Methodology, Investigation, Validation, Writing - review and editing

AS: Conceptualization, Methodology, Investigation, Validation, Writing—original draft

PM: Conceptualization, Methodology, Investigation, Validation, Writing - review and editing

BCO: Methodology, Investigation, Formal analysis, Validation, Software, Writing - review and editing.

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